

Monthly Intelligence Report

Edition 01 — The Paks Nuclear Corridor

Where four VVER-440 units operating + two VVER-1200 units under construction
concentrate ~45-50% of Hungarian baseload on a single Danube-bank site. Inaugural
edition · July 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

3,502

MAVIR + MVM/E.ON heritage

GRID LINES MAPPED

~4,261

400 / 220 / 120 kV + MV

MC SIMULATIONS

35.02 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). For Hungary, its engine processes 74,000+ source data points per run, executes 35.02 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 830+ million arithmetic operations — producing 210,000+ output data points with full uncertainty quantification across 3,502 substations and ~4,261 grid lines.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources — MAVIR network statement, MEKH quality reports, MVM + E.ON DSO filings, KSH RegDat, OMSZ meteorology, OVF Danube/Tisza hydrology, OAH nuclear oversight, GovCERT-Hungary + SZTFH cyber posture — making the methodology fully auditable and reproducible. Hungary is the third and largest member of the CEE-South-2026 cohort, deployed in triple-drop with Slovenia and Slovakia on the 9 July 2026 anchor.

Substation in Focus

Beremend alállomás

Baranya, HU231 · 132 kV

0.447

R_FINAL

Medium

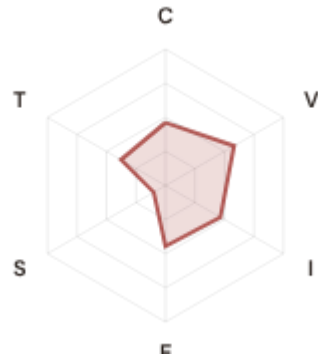
CLASSIFICATION

0.101

CI WIDTH

84th

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.461 / 0.30 (154%)

I Infrastructure: 0.465 / 0.25 (186%)

S Saturation: 0.100 / 0.20 (50%)

V Voltage: 0.582 / 0.10 (582%)

E Economic: 0.447 / 0.10 (447%)

T Transition: 0.377 / 0.05 (754%)

MODIFIERS

R3 Consequence: 1.044 ↑ amplifies

R4 Graph Criticality: 1.003 → neutral

R6a Restoration: 1.075 ↑ amplifies

R6b Seismic: 1.022 ↑ amplifies

R7 Digital Readiness: 0.980 ↓ dampens

This month's spotlight — automatically selected for June 2026 — is **Beremend alállomás** in Baranya, HU231. With an R_final of 0.447 (Medium band, 84th fleet percentile), its risk profile is dominated by the T (Transition) component at 754% of its weight ceiling, followed by V (Voltage) at 582%. Modifiers collectively shift R_base by 32.1%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Hungary's Act LXIX/2024 NIS2 transposition + SZTFH competent-authority designation require.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

46

High/Critical + R7 < median

AVG R7 MODIFIER

0.9795

Range [0.99, 1.03]

HIGH-CONSEQUENCE SUBS

923

26.4% · R3 ≥ 1.04

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

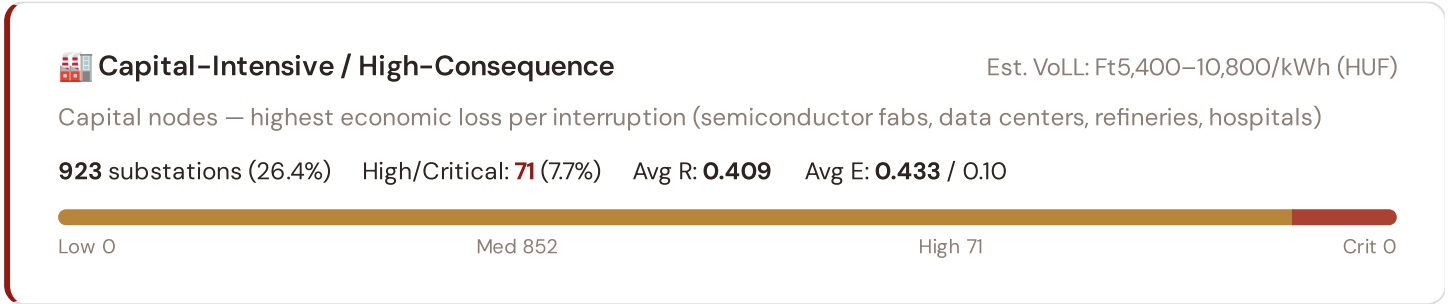


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **46 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 0.9794$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **HU311 (7)**, **HU233 (7)**, **HU120 (5)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.





Industrial / Medium–Large Enterprise

Est. VoLL: Ft2,700–5,400/kWh (HUF)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

833 substations (23.8%) High/Critical: **43** (5.2%) Avg R: **0.392** Avg E: **0.431** / 0.10



Commercial / SME–Dense

Est. VoLL: Ft1,080–2,700/kWh (HUF)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

873 substations (24.9%) High/Critical: **28** (3.2%) Avg R: **0.385** Avg E: **0.438** / 0.10



Light / Agricultural / Rural

Est. VoLL: Ft360–1,080/kWh (HUF)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

873 substations (24.9%) High/Critical: **6** (0.7%) Avg R: **0.363** Avg E: **0.437** / 0.10

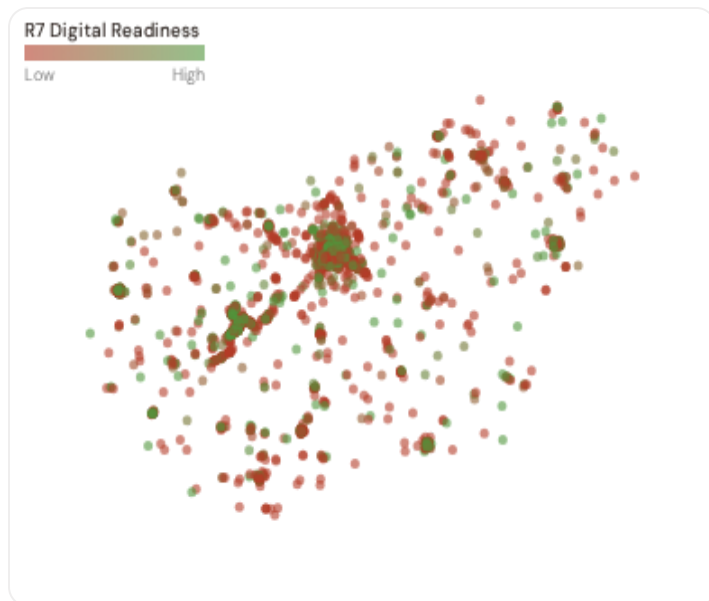


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **71 substations** combine capital-intensive economic fabric ($R3 \geq 1.04$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **HU110 (450)**, **HU213 (90)**, **HU120 (85)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 5.7%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 NUTS-3 Megye Digital Readiness Ranking

Megyék ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions + GovCERT–Hungary maturity). Cross-referenced with average R_{median} to identify megyék combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 NUTS-3 REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Bács-Kiskun ▲ high R	0.9672
2	Baranya ▲ high R	0.9699
3	Somogy ▲ high R	0.9710
4	Komárom-Esztergom ▲ high R	0.9746
5	Jász-Nagykun-Szolnok ▲ high R	0.9750
6	Hajdú-Bihar ▲ high R	0.9768
7	Fejér ▲ high R	0.9771
8	Borsod-Abaúj-Zemplén ▲ high R	0.9778
9	Veszprém ▲ high R	0.9788
10	Pest ▲ high R	0.9790
11	Vas ▲ high R	0.9797
12	Tolna ▲ high R	0.9797
13	Budapest	0.9805
14	Zala ▲ high R	0.9821
15	Heves ▲ high R	0.9826

NUTS-3-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Bács-Kiskun** has the lowest average R7 (0.9672), while **Békés** leads at 0.9921 — a spread of 0.0249 across the R7 range. **10 nuts-3 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9795, median = 0.9794; 0 substations classified HIGH cyber-exposure; 46 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **18 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

18

95 variables

WEEKLY / MONTHLY

3

High-frequency feeds

QUARTERLY / ANNUAL

10

Regular refresh cycle

STATIC / DERIVED

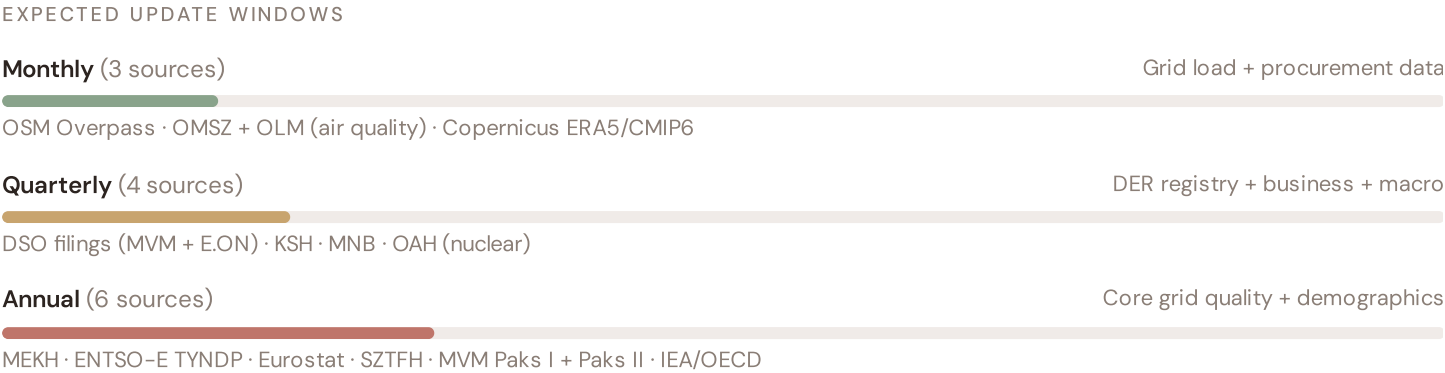
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Standards + scenarios

Data Source Registry — June 2026

d01_mavir	MAVIR — Magyar Villamosenergia-ipari Átviteli Rendszerirányító (TSO)	DAILY	C2,C3,I1,I2,I3,R4	14 vars
d03_omsz	OMSZ — Országos Meteorológiai Szolgálat	DAILY	I1,I2,I3,R6_climate	8 vars
d03b_ovf	OVF — Országos Vízügyi Főigazgatóság (Hungarian Water Authority)	DAILY	R6c_flood	6 vars
d14_hupx	HUPX — Hungarian Power Exchange (NEMO) + CORE FB-MC	DAILY	T1,E1	4 vars
d05_osm	OSM Overpass — grid topology (ISO3166- 1=HU area filter)	MONTHLY	I1,I2,I3,R4	8 vars
d06_aqi	OMSZ + OLM — Hungarian Air Quality Network (ISO 9223 corrosion)	MONTHLY	R5_corrosion,I8	5 vars
d07_copernicus	Copernicus ERA5 + CMIP6	MONTHLY	R6_climate	4 vars
d09_govcert_hu	GovCERT-Hungary (National Cyber Defence Institute, SZTFH lineage)	CONTINUOUS	R7_cyber	5 vars
d02b_dso	6 regional DSOs (MVM Démász/Émász/Elmű + E.ON Észak/Dél/Tiszántúli)	QUARTERLY	C1,V1,V2,I2,I3	18 vars
d04_ksh	KSH — Központi Statisztikai Hivatal	QUARTERLY	E1,E2,S1,S2,S3	16 vars
d04b_mnb	MNB — Magyar Nemzeti Bank (central bank · ESCB non-€)	QUARTERLY	E1,E2	4 vars
d08_oah	OAH — Országos Atomenergia Hivatal (nuclear safety)	QUARTERLY	R6_seismic,I3,I4	5 vars
d02_mekh	MEKH — Magyar Energetikai és Közmű- szabályozási Hivatal	ANNUAL	C1,C2,V1,V2,T1	12 vars
d10_sztfh	SZTFH — Szabályozott Tevékenységek Felügyeleti Hatósága (NIS2 CA)	ANNUAL	R7_cyber	3 vars
d11_entsoe	ENTSO-E TYNDP + Transparency Platform	ANNUAL	I1,I2,T1	6 vars
d12_eurostat	Eurostat — EU-27 NUTS-3 benchmarks	ANNUAL	E1,S1,T1	8 vars
d13_mvm_paks	MVM Paksi Atomerőmű (Paks I) + Paks II Atomerőmű Zrt.	ANNUAL	I1,I3,R6_seismic	5 vars
d15_iea_oecd	IEA + OECD — energy benchmarks (HU joined OECD 7 May 1996, 24th member)	ANNUAL	T1,E1	6 vars

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **3,502 substation scores remain unchanged** this edition. The fleet median R of 0.383 (mean 0.388) reflects a distribution skewed toward the lower bands, with 0.2% in Low and 95.6% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

9 changes tracked across PO, PI, and P2 releases

HU-S29-9	ENHANCED	Greenfield thin-shell on post-Slovakia-hotfix architecture — 8 pages authored directly on CountryRenderer.Safe + normalizeMeta() (KB §68.10/11)	KB §69
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HU-S29-8	NEW	Section G deepDives rotation — 12 HU megyék anchored on Paks I+II nuclear corridor (Tolna)	intelligence G
HU-S29-7	ENHANCED	Edition patcher anchored at FIRST_REFRESH 2026-07-09 (CEE-South-2026 triple-drop with Slovenia + Slovakia)	intelligence
HU-S29-6	NEW	ESG report — hungary entry in COUNTRY_SOURCES (14 HU-specific references including Danube/Tisza flood 2013/2024 + Paks I+II + GovCERT-HU)	esg-report
HU-S29-5	NEW	R3 5-tier calibration — Capital / Industrial-Major / Industrial-Secondary / Light-Rural / East-Lagging (vs SK 4-tier; reflects sharp Budapest-vs-Nógrád gradient)	methodology
HU-S29-4	NEW	R6 modifier set — Danube/Tisza flood HIGH (2013 Budapest 891 cm record) + Pannonian heatwave + Komárom-Mór seismic axis	methodology
HU-S29-3	DATA	d05_osm LIVE — 3,502 substations + 4,261 power lines via ISO3166-1=HU area filter (7-border bbox-bleed protection — leakiest cohort)	methodology
HU-S29-2	DATA	6 regional DSOs (MVM Déomász/Éomász/Elmű + E.ON Észak/Dél/Tizántúli) + MAVIR TSO + MEKH regulator + OAH nuclear safety wired	methodology
HU-S29-1	ENHANCED	KB v25 §69 — Hungary inaugural onboarding (CEE-South-2026 cohort member 3 of 3 — cohort COMPLETE; HUF currency, NOT eurozone)	KB §69

SECTION D — MONTHLY DEEP-DIVE

Tolna megye — The Paks Nuclear Corridor

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation covers Hungary's most material NUTS-3 narratives: **Ed. 01** Tolna — Paks I+II nuclear corridor · **02** Budapest — capital load + 2013 Danube flood · **03** Győr-Moson-Sopron — Audi automotive backbone · **04** Borsod-Abaúj-Zemplén — Miskolc heavy industry + Tiszaújváros chemical · **05** Baranya — Mecsek seismic + Pécs services pole · **06** Hajdú-Bihar — Debrecen BMW ramp-up · **07** Bács-Kiskun — Kecskemét Mercedes + Pannonian heat · **08** Békés — east-lagging + RO border · **09** Veszprém — Bakony R&D + chemical · **10** Csongrád-Csanád — south borders + Szeged · **11** Szabolcs-Szatmár-Bereg — Tisza floods + UA border · **12** Heves — Mátra coal exit + Eger.

D.1 Why Tolna megye, Why Now

CORRIDOR SUBSTATIONS

111

12 EHV · 99 HV

HIGH BAND

14

12.6% of corridor

CORRIDOR MEDIAN R

0.416

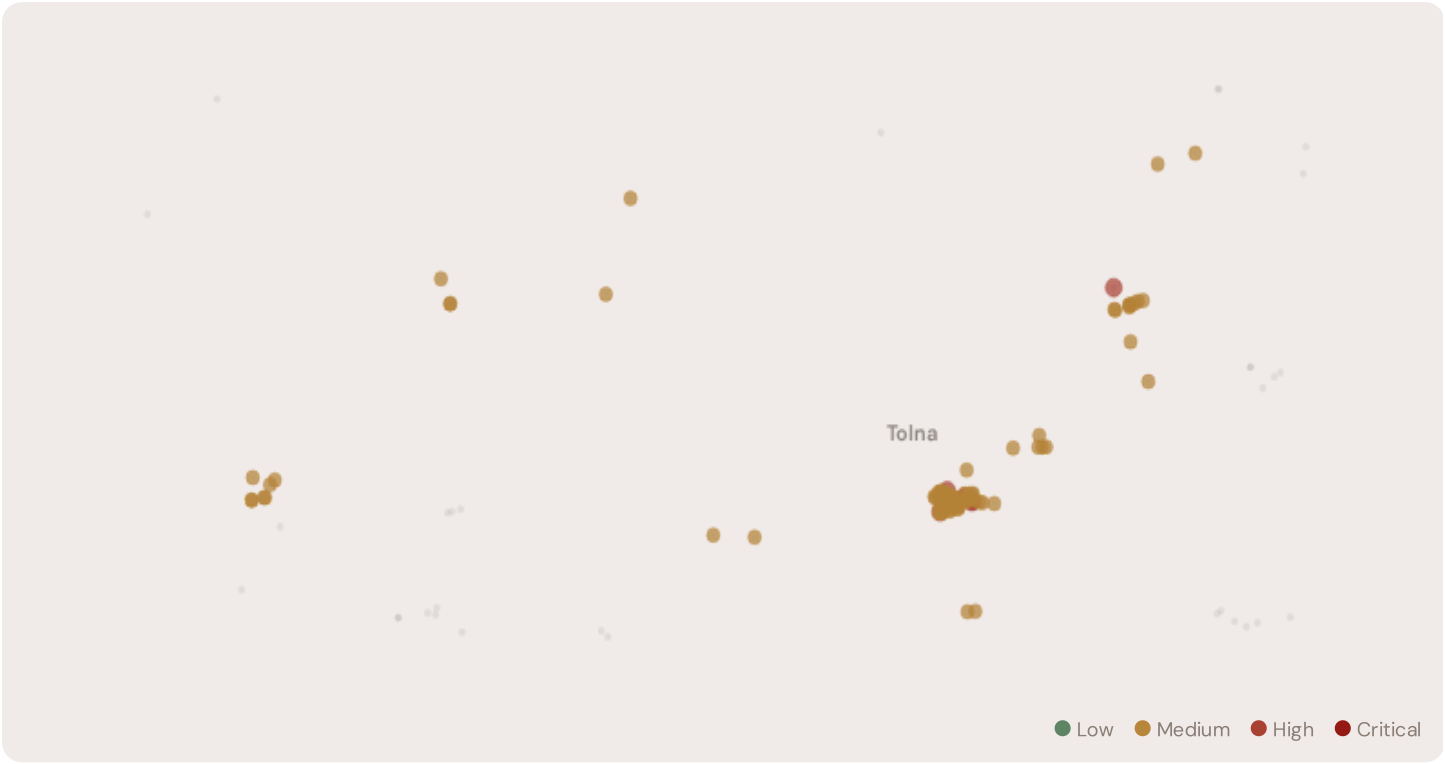
Fleet median: 0.383

WORST NUTS-3 MEGYE

Tolna

The Tolna megye — Paks I+II nuclear corridor hosts **111 substations** across 1 nuts–3 regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **14 substations (12.6%)** are classified High, with only **0** in the Low band. 74% of corridor substations score above the national fleet median of 0.383. The corridor median R of 0.416 reflects the local risk profile. Tolna megye (HU233) is Hungary's single-site nuclear concentration zone. All four Paks I VVER-440/V-213 units operate at a single site on the Danube's right bank, ~100 km south of Budapest — Units 1–4 commissioned 1982–1987, ~500 MWe each post-uprate, combined ~1,916 MWe generating ~45–50% of Hungarian electricity. Paks II Units 5+6 (VVER-1200/V-491, ~1,200 MWe each) are co-located ~500 m from Paks I; Unit 5 First Concrete was poured 5 February 2026 by Rosatom — the first new-NPP first-concrete in any EU country since Hinkley Point C Unit 2 (Dec 2019), and the first Russian-built NPP in an EU member state. Target operation ~2031–2032 for Unit 5, ~2032–2033 for Unit 6. US OFAC sanctions on the project were lifted in Dec 2025. The Light-Rural R3 tier (1.06) reflects Tolna's predominantly agricultural character beyond the nuclear site, but the single-site concentration drives elevated R3 + R6_seismic + R6c_flood (Danube right-bank) modifier sensitivity at the corridor level. OAH safety oversight, ENTSO-E reserve obligations and MAVIR balancing cascade across the E.ON Dél-dunántúli DSO simultaneously in any Paks trip scenario. When Paks II Units 5+6 come online, HU nuclear share could approach 70%+ — among the world's highest.

D.2 Corridor Map and Scoring

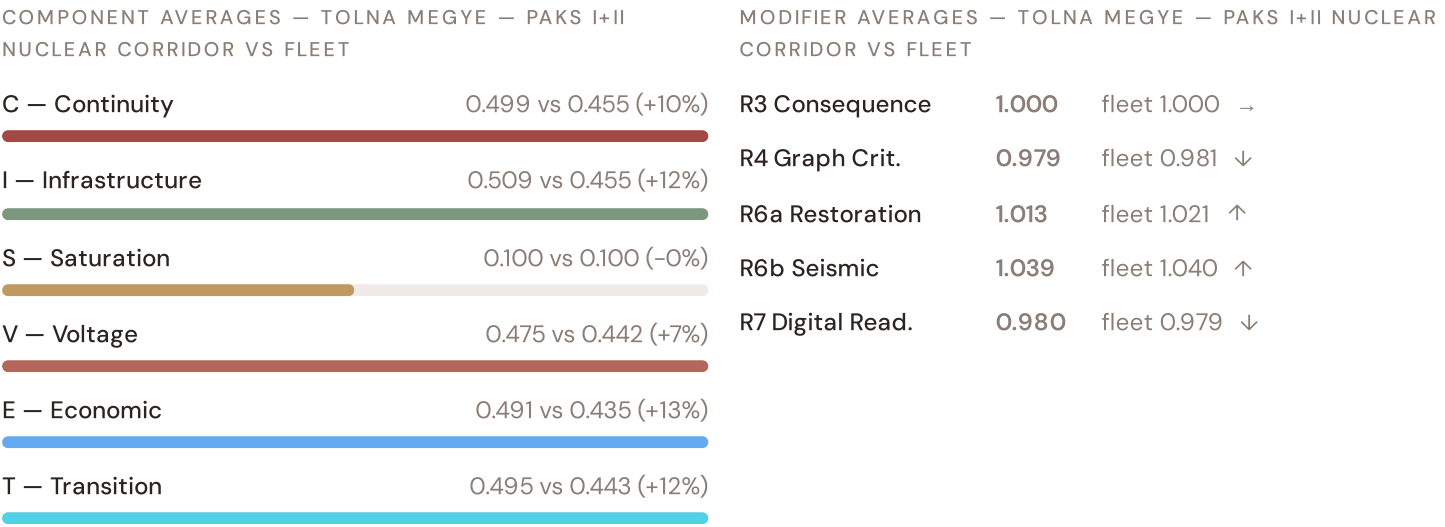


SUBSTATION	MEGYE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Paks DÉDÁSZ állomás	Tolna	0.593	High	0.666	0.577	0.520	0.571	0.100	0.484	C
HU_102451	Tolna	0.593	High	0.640	0.571	0.645	0.437	0.100	0.606	
HU_102394	Tolna	0.579	High	0.669	0.464	0.643	0.568	0.100	0.603	C

SUBSTATION	MEGYE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
HU_102435	Tolna	0.533	High	0.557	0.681	0.591	0.448	0.100	0.445	V
HU_102414	Tolna	0.525	High	0.635	0.457	0.557	0.511	0.100	0.561	
HU_102450	Tolna	0.515	High	0.385	0.682	0.629	0.534	0.100	0.613	V
HU_102526	Tolna	0.515	High	0.661	0.472	0.484	0.489	0.100	0.402	C
19908	Tolna	0.512	High	0.610	0.399	0.526	0.529	0.100	0.631	
HU_102530	Tolna	0.509	High	0.621	0.622	0.410	0.568	0.100	0.549	
Szekszárd állomás	Tolna	0.508	High	0.650	0.518	0.609	0.395	0.100	0.563	

... 101 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis



The dominant risk driver across the Tolna megye — Paks I+II nuclear corridor corridor is **T (Transition)**, averaging 0.495 against a fleet average of 0.443 — occupying 990% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.491 vs fleet 0.435 (491% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.000 (fleet: 1.000), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.039, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 NUTS-3 Megye Breakdown



The nuts-3 regions-level breakdown reveals significant intra-regional variation. **Tolna** leads with a mean R of 0.423 across 111 substations, while **Tolna** scores 0.423 — a spread of 0.000 within a single corridor. The SSI's substation-level

granularity reveals that the Tolna megye — Paks I+II nuclear corridor is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

0 Tolna megye — Paks I+II nuclear corridor substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Tolna sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

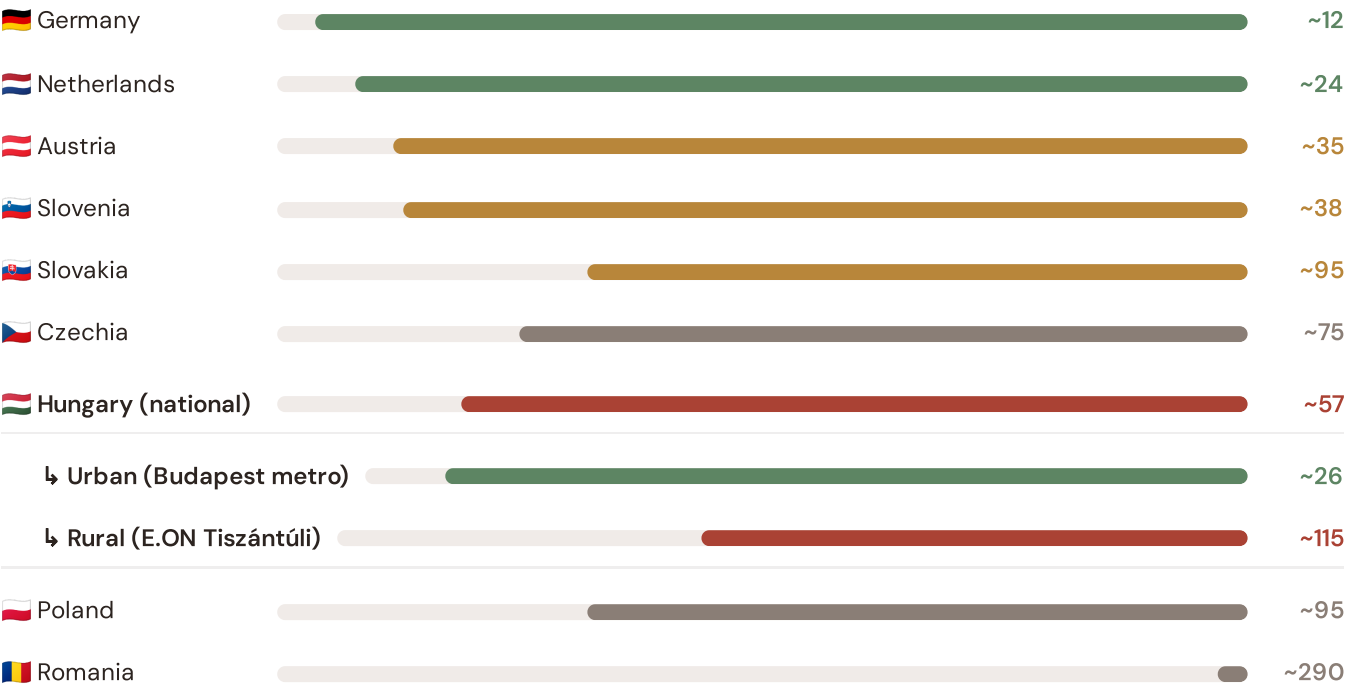
SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Paks-corridor analysis hinges on Hungary's continuity performance relative to European peers, especially the Visegrád-4 + CEE-South cohort siblings. Future editions rotate through: VoLL comparison (Ed. 02), DER saturation vs Germany (03), CMIP6 climate hazard vs DACH peers (04), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER 7th Benchmarking Report (2022); MEKH 2023 quality reports; per-DSO filings (MVM + E.ON 2023). Hungary sub-categories from MEKH territory classification (Budapest metro vs E.ON eastern rural Tiszántúli). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

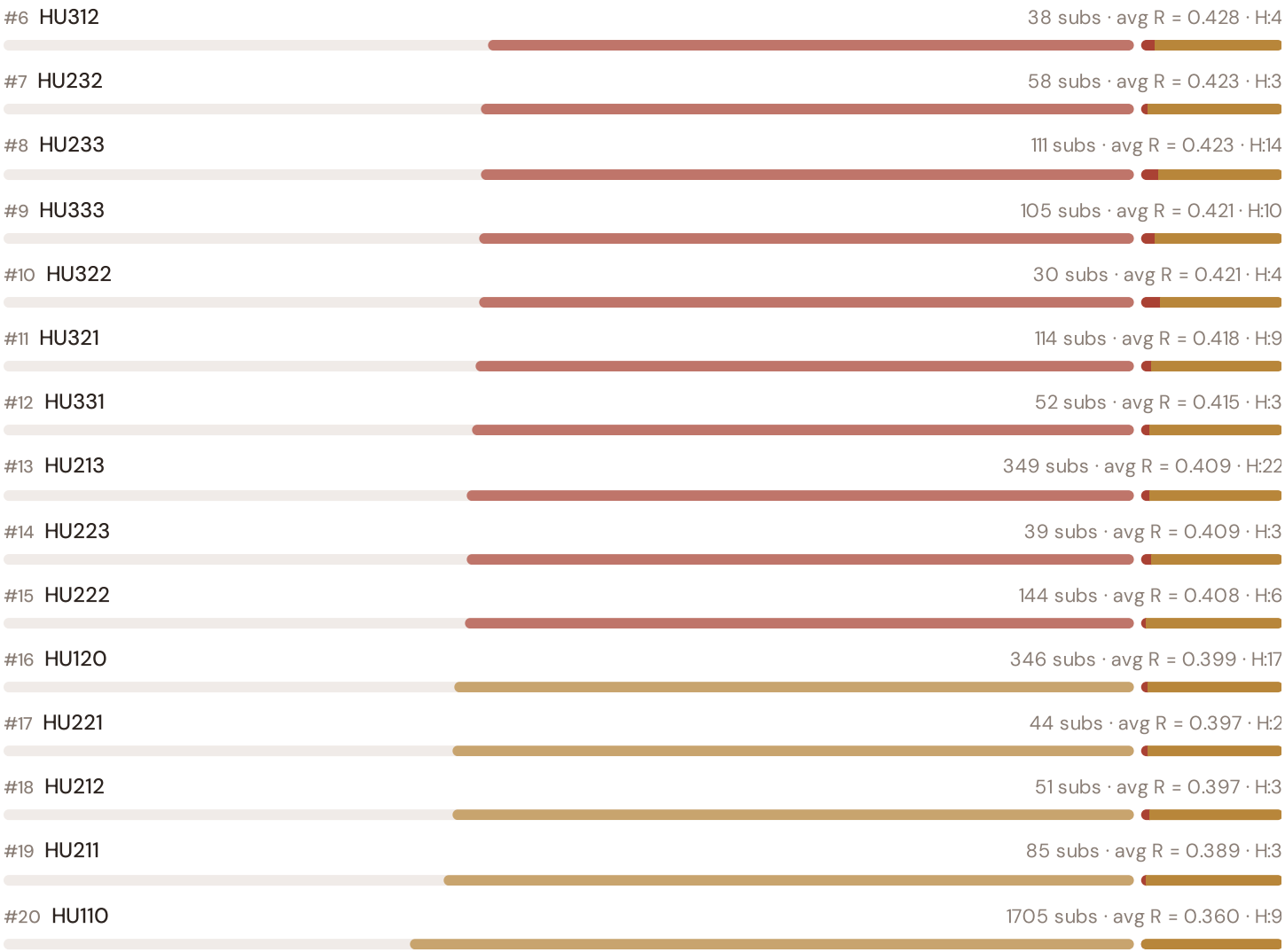
This month's key insight: The Tolna megye — Paks I+II nuclear corridor corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0x the fleet average** of 0.455. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 20 Hungarian NUTS-3 Megyék

Where does Tolna megye sit within the national landscape? Megyék sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



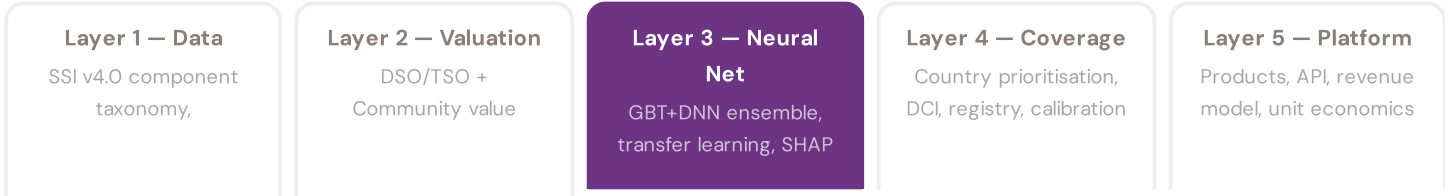


The Tolna megye — Paks I+II nuclear corridor ranks **#0 of 20 nuts–3 regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are HU332, HU323, HU313, while the three most resilient are HU110, HU211, HU212. 19 of 20 score above the fleet average of 0.388. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to MVM/E.ON DSOs and to MAVIR — and to the community?"* Each edition explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 3,502 substations with complete feature vectors. Average CI_width of 0.068 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 6 Low, 3348 Medium, 148 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Szeged & South Borders

Deep-dive into Csongrád-Csanád megye's grid risk profile — substation map, component analysis, nuts-3 regions breakdown. European Context Card: Szeged + electronics; Csongrád-Csanád name 2020 Trianon centenary restoration; RS/RO border.

NEXT DATA REFRESH

Q3 2026 — 4 Quarterly Sources

DSO filings (MVM + E.ON), KSH, MNB, OAH (nuclear) are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **MEKH — Magyar Energetikai és Közmű-szabályozási Hivatal** (12 variables → C1,C2,V1,V2,T1).

FLEET SPOTLIGHT

Szabolcs-Szatmár-Bereg: Highest Risk Concentration

Szabolcs-Szatmár-Bereg has the highest concentration of High/Critical substations: **7 of 33 (21%)** are in the upper risk bands — avg R = 0.453. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Csongrád-Csanád megye**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Hungarian utility, MEKH, MAVIR, MVM, E.ON Hungary, OAH, SZTFH, GovCERT-Hungary, or an academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Hungary Edition 01 (Inaugural)

Published 9 July 2026 · SSI Index v4.0.2 · 3,502 substations · ~4,261 grid lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5

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